



Operations Research

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Sample solution to the demo tasks 5

Demo Exercises

Exercise D5.1. *Organic Farmer*

Problem

Organic farmer Georgina is planning her crop cultivation for the next 15 years. Using detailed statistical analyses, she has forecasted the following expected profits:

	Profit per hectare	
	With good soil (g)	With bad soil (b)
Wheat (W)	10	5
Corn (C)	12	4
Potatoes (P)	15	6
Rye (R)	7	7

The computer-savvy farmer wants to use an IP solver to determine a cultivation plan that maximizes her total profit over the next 15 years. She introduces variables $x_{p,q,t} \geq 0$, where $p \in P := \{W, C, P, R\}$ denotes the product, $q \in Q := \{g, b\}$ indicates the soil quality and $t \in T := \{1, 2, \dots, 15\}$ represents the year. The variable $x_{p,q,t}$ therefore describes the quantity of product p cultivated on land of quality q in year t .

- Specify the objective function to be maximized as a function of $x_{p,q,t}$.
- Formulate the following constraints as linear (in)equations in the variables $x_{p,q,t}$. You may introduce additional *binary* variables if required.
 - No more than 1000 hectares of land may be used for crop cultivation every year.
 - If an agricultural product is to be cultivated in year t , and it was cultivated in each of the two preceding years on at most 50 hectares, then it may be cultivated on good soil in year t ; otherwise, it can only be grown on poor soil. Assume that no crops were cultivated before year $t = 1$.
 - The total area of potatoes cultivated over all years may not exceed 1500 hectares.
 - If rye has been cultivated in 3 consecutive years, no rye may be cultivated in the following year.

- (v) If wheat and rye are cultivated in a year, at least one other agricultural product must be cultivated on an area of at least 300 hectares.

Solution

- a) The objective function to be maximized is given by

$$\sum_{t \in T} 10x_{W,g,t} + 5x_{W,b,t} + 12x_{C,g,t} + 4x_{C,b,t} + 15x_{P,g,t} + 6x_{P,b,t} + 7(x_{R,g,t} + x_{R,b,t}).$$

- b) (i)

$$\sum_{p \in P} \sum_{q \in Q} x_{p,q,t} \leq 1000 \quad \forall t \in T$$

- (ii) Introduce additional binary variables $A_{p,t} \in \{0, 1\}$ with $p \in P, t \in T$ that describe whether a crop is grown on more than 50 hectares in a given year. Obtain the constraints

$$\begin{aligned} 950A_{p,t} &\geq (x_{p,g,t} + x_{p,b,t}) - 50 && \forall p \in P, t \in T \\ x_{p,g,t} &\leq 1000(1 - A_{p,t-1}) && \forall p \in P, t \in \{2, \dots, 15\} \\ x_{p,g,t} &\leq 1000(1 - A_{p,t-2}) && \forall p \in P, t \in \{3, \dots, 15\} \end{aligned}$$

- (iii)

$$\sum_{q \in Q} \sum_{t \in T} x_{P,q,t} \leq 1500$$

- (iv) We introduce binary variables $R_t \in \{0, 1\}$ for $t \in T$ that describe whether rye is grown in year t :

$$1000R_t \geq x_{R,g,t} + x_{R,b,t} \quad \forall t \in T$$

Then we can write the constraint as

$$x_{R,q,t} \leq 1000(3 - R_{t-3} - R_{t-2} - R_{t-1}) \quad \forall q \in Q, t \in \{4, \dots, 15\}.$$

- (v) We introduce binary variables $W_t \in \{0, 1\}$, $t \in T$, which indicate whether wheat is grown in year t . Furthermore, we introduce variables $P_t \in \{0, 1\}$ for $t \in T$, which indicate whether at least 300 hectares of potatoes are grown in year t .

$$\begin{aligned} 1000W_t &\geq x_{W,g,t} + x_{W,b,t} && \forall t \in T \\ 300P_t &\leq x_{P,g,t} + x_{P,b,t} && \forall t \in T \\ x_{C,g,t} + x_{C,b,t} &\geq 300(W_t + R_t - 1 - P_t) && \forall t \in T \end{aligned}$$

Exercise D5.2. School Planning

Problem

Welcome to *Newville* – a city so new it still smells like fresh paint. The city planners are deciding which schools to build. There are $I = \{1, \dots, 15\}$ possible school sites, each

with a capacity of $c_i \in \mathbb{R}_{>0}$ and a construction cost of $f_i \in \mathbb{R}_{>0}$ Euros. The city has $J = \{1, \dots, 800\}$ students who need to be assigned to schools. Each student $j \in J$ has a preference score $s_{i,j} \in \mathbb{R}_{>0}$ for each school site $i \in I$, where $s_{i,j} > s_{i',j}$ indicates that student j prefers school site i over i' . Your task is to help the city planners decide which schools to build and how to assign students to schools.

Let $y_i \in \{0, 1\}$ be a variable which is 1 if school $i \in I$ is built and 0 otherwise. Also, let $x_{i,j} \in \{0, 1\}$ be a variable which is 1 if student $j \in J$ is assigned to school $i \in I$ and 0 otherwise.

Formulate the following constraints as linear (in-)equalities. You may introduce additional variables. If you do so, please also state their intuitive meaning in words:

- a) Each student must be assigned to exactly one school.
- b) The total number of students assigned to each school cannot exceed the school's capacity.
- c) A student may be assigned to a school only if the school is built.
- d) Let the parameter $d_{i,i'} \in \mathbb{R}_{>0}$ denote the distance (in kilometers) between two school sites $i, i' \in I$ with $i \neq i'$. The distance between any two built schools must be at least 5 km.
- e) Assume that the city is divided into two zones, A and B . The parameter $A_i \in \{0, 1\}$ indicates whether school site $i \in I$ is located in zone A ($A_i = 1$) or in zone B ($A_i = 0$). If more than 200 students are assigned to schools in zone A , then at least 300 students must be assigned to schools in zone B .
- f) Formulate a single objective function that minimizes total construction costs while maximizing total preference scores of students.

Solution

- a) $\sum_{i \in I} x_{i,j} = 1 \quad \forall j \in J$
- b) $\sum_{j \in J} x_{i,j} \leq c_i \quad \forall i \in I$
- c) $x_{i,j} \leq y_i \quad \forall i \in I, j \in J$
- d) We introduce an auxiliary variable $z_{i,i'} \in \{0, 1\}$: both schools i and i' are open $\implies z_{i,i'} = 1$:

$$1 + z_{i,i'} \geq y_i + y_{i'} \quad \forall i, i' \in I, i \neq i'$$

Then we need the constraint:

$$d_{i,i'} z_{i,i'} \geq 5 z_{i,i'}$$

- e) We introduce an auxiliary variable $y \in \{0, 1\}$ such that $y = 1 \iff$ more than 200 students are assigned to zone A :

$$\sum_{i \in I} \sum_{j \in J} A_i x_{i,j} \leq 200 + My$$

$$\sum_{i \in I} \sum_{j \in J} A_i x_{i,j} \geq 201y$$

Then we need the constraint:

$$\sum_{i \in I} \sum_{j \in J} (1 - A_i) x_{i,j} \geq 300y$$

$$\text{f) } \min \sum_{i \in I} f_i y_i - \sum_{i \in I} \sum_{j \in J} s_{i,j} x_{i,j}$$